

Pre-Project SWOT Analysis:

Purpose

The purpose of this project is to identify and map high carbon emission hotspots in urban areas, such as busy traffic intersections and industrial zones. By analyzing these locations, the project aims to raise awareness about urban carbon emissions and suggest practical solutions to reduce the carbon footprint and improve urban environmental sustainability.

Project Overview

- **Project Name:** Urban Emission Hotspot Mapping
 - **Project Objective / Goal:** Identify high-emission zones in urban areas using GIS mapping, open data, and community inputs, and propose targeted solutions to reduce carbon emissions and improve urban sustainability.
 - **Project Owner / Team:** Team Members:
 1. Shadhana .S
 2. Srinith .R
 3. Sindhiya .S.T
 4. Varnikha Sri .S
 - **Planned Start Date:** March 7, 2026
 - **Key Stakeholders:**
 1. Municipal Corporation / Urban Planning Department
 2. Local Communities and Citizens
 3. Environmental NGOs
 4. Transport Authorities
 5. Academic Mentors
-

SWOT Analysis

Strengths (*Internal, Positive Factors*):

- **Multidisciplinary team skills** combining data science, sustainability knowledge, and mapping technologies.
- **Availability of open-source data** such as traffic data, satellite imagery, pollution datasets, and weather data.
- **Use of GIS and Python** tools, which allow efficient visualization and analysis of environmental data.
- **Citizen science approach** enabling community members to contribute real-world observations about pollution hotspots.
- **Low-cost technology options** such as open-source mapping platforms and affordable sensors.

Weaknesses (*Internal, Negative Factors*):

- **Limited access to real-time emission sensors**, which may reduce data accuracy.
- **Skill gaps in advanced GIS or environmental modelling** among team members.
- **Time constraints (35–40 hours)** may limit depth of analysis.
- **Dependence on secondary datasets**, which may not always be updated or location specific.
- **Small team size** may limit large-scale data collection across multiple areas.

■ Opportunities (*External, Positive Factors*)

- *Municipal governments seeking data-driven urban planning solutions.*
- *Availability of new tools such as open-source GIS platforms, satellite pollution data, and AI-based analytics.*
- *Potential collaborations with environmental NGOs and civic groups.*
- *Opportunity to build an interactive dashboard that could be used for future research or smart-city initiatives.*
- *Growing public awareness of air pollution, encouraging community participation.*
- *Increasing global focus on climate action and carbon reduction, making the project highly relevant.*

■ Threats (*External, Negative Factors*):

- *Incomplete or inaccurate public datasets affecting analysis results.*
- *Rapid changes in urban infrastructure or traffic patterns, making data outdated quickly.*
- *Limited engagement from local communities or authorities.*
- *Technical issues with mapping software or sensor integration.*
- *Competing research projects or lack of attention from policymakers.*

Next Steps / Strategic Actions

Leverage Strengths:

- **Use open-source GIS platforms and Python libraries to create accurate emission maps.**
- **Combine team members' skills in coding, mapping, and sustainability analysis.**

Address Weaknesses:

- **Use publicly available pollution datasets and traffic data to supplement sensor data.**
- **Divide tasks clearly among team members (data collection, modeling, visualization, reporting).**

Act on Opportunities:

- **Build an interactive heatmap dashboard to visualize emission hotspots clearly.**
- **Engage local citizens through surveys or simple reporting tools for pollution observations.**
- **Share project findings with municipal authorities or sustainability organizations.**

Plan for Threats:

- **Cross-verify data from multiple sources to improve reliability.**
- **Keep the project focused on a specific urban area to ensure realistic results within the timeframe.**
- **Maintain backup datasets and tools to avoid technical disruptions.**